Hall-Littlewood Symmetric Functions

0. Introduction.

Let $x_1, \dots, x_n, t$ be commuting variables over $\mathbb{Z}$. The Hall-Littlewood polynomials $P_\lambda(x_1, \dots x_n, t)$ are symmetric functions which interpolate between the Schur polynomials s_λ and the monomial symmetric function m_λ:

$$P_\lambda(x_1, \dots x_n, 0) = s_\lambda(x_1, \dots x_n)$$

$$P_\lambda(x_1, \dots, x_n, 1) = m_\lambda(x_1, \dots, x_n).$$

Moreover, they form a $\mathbb{Z}[t]$-basis of $\Lambda[t]$ and we have:

$$s_\lambda(x_1, \dots, x_n) = \sum_\mu K_{\lambda\mu}(t)\, P_\mu(x_1, \dots x_n, t),$$

so that the above is a t-analogue of the classical expression:

$$s_\lambda(x) = \sum_\mu K_{\lambda\mu}\, m_\mu,$$

where $K_{\lambda\mu}(t)$ are the so called Kostka-Foulkes polynomials.

1. The polynomials R_λ.

Let λ be a partition with $\ell(\lambda) \leq n$. Define:

$$R_\lambda(x_1, \dots, x_n; t) = \sum_{w \in S_n} w \cdot \left(x_1^{\lambda_1} \cdots x_n^{\lambda_n} \cdot \prod_{i<j} \frac{x_i - t x_j}{x_i - x_j} \right).$$

Ex. $R_{\square\square}(x_1, x_2; t) = x_1^2 + x_1 x_2 + x_2^2 - t x_1 x_2$.

Since $a_\delta = \prod_{i<j} (x_i - x_j)$, we rewrite

$$R_\lambda(x_1, \dots, x_n; t) = \frac{1}{a_\delta} \sum_{w \in S_n} \operatorname{sgn}(w)\, w \cdot \left(x^\lambda \prod_{i<j} (x_i - t x_j) \right) \quad (*)$$

Prop. i) $R_\lambda(x; t)$ is a hom. symmetric polynomial in $x_1, \dots, x_n$ of degree $|\lambda|$ with coeff. in $\mathbb{Z}[t]$.

ii) $R_\lambda(x; t) = \sum_\mu u_{\lambda\mu}(t)\, s_\mu(x)$, where

$u_{\lambda\mu}(t) = 0$ unless $|\lambda| = |\mu|$ and $\mu \trianglelefteq \lambda$. Moreover,

$$u_{\lambda\lambda}(t) = \prod_{i \geq 0} [m_i]_t!$$

with $m_i = \#$ parts of λ equal to i, $[n]_t = 1 + t + t^2 + \cdots + t^{n-1}$.

iii) $R_\lambda(x; 0) = s_\lambda(x)$, $R_\lambda(x; 1) = |S_n^\lambda|\, m_\lambda(x)$.

Proof.

i) The sum in the RHS of (*) is anti-symmetric in x_i's, whence it is divisible by a_δ in $\mathbb{Z}[x_1, \dots, x_n, t]$. Thus, the RHS of (*) is a hom. symmetric polynomial with coeff. in $\mathbb{Z}[t]$.

iii) We have $R_\lambda(x,0) = \frac{1}{a_\delta} \sum_{w \in S_n} \operatorname{sgn}(w)\, w \cdot x^{\lambda+\delta} = \frac{a_{\lambda+\delta}}{a_\delta} = s_\lambda$.

Also, $R_\lambda(x,1) = \sum_{w \in S_n} w \cdot x^\lambda = |S_n^\lambda|\, m_\lambda$.

ii) Any of the summands in $\prod_{i<j} (x_i - t x_j)$ is of the form:

$$\prod_{i<j} x_i^{r_{ij}} (-t x_j)^{r_{ji}}$$

for some $n \times n$ matrix (r_{ij}) of 0's and 1's such that:

$$r_{ii} = 0, \quad r_{ij} + r_{ji} = 1, \quad i \neq j.$$

Hence, we have:

$$x^\lambda \cdot \prod_{i<j} x_i^{r_{ij}} (-t x_j)^{r_{ji}} = (-t)^d x^\alpha$$

where $\alpha = \lambda_i + \sum_j r_{ij}$ and $d = \sum_{i<j} r_{ji}$.

Hence, R_λ is a sum of terms of the form:

$$(-t)^d a_\alpha \cdot a_\delta^{-1}.$$

Since $a_\alpha = 0$ if α has repetitions, we may assume that

$$\alpha_{w(1)} > \alpha_{w(2)} > \cdots > \alpha_{w(n)} \geq 0 \quad \text{for some } w \in S_n.$$

Thus, $w \cdot \alpha = \mu + \delta$ for some partition μ, whence

$$a_\alpha \cdot a_\delta^{-1} = \operatorname{sgn}(w)\, s_\mu.$$

We now show that $\mu \trianglelefteq \lambda$. Set (s_{ij}) to be the $n \times n$ matrix

$$s_{ij} = r_{w(i), w(j)}$$

so that $s_{ii} = 0$, $s_{ij} + s_{ji} = 1$ $(i \neq j)$, and

$$\mu_i + n - i = \alpha_{w(i)} = \lambda_{w(i)} + \sum_j s_{ij}.$$

Thus, $\mu_1 + \cdots + \mu_k = \lambda_{w(1)} + \cdots + \lambda_{w(k)} + \sum_{i=1}^{k} \sum_{j=1}^{n} s_{ij} - \sum_{i=1}^{k} (n-i)$.

In particular, $|\mu| = |\lambda|$.

Now, we have:

$$\sum_{i=1}^{k} \sum_{j=1}^{n} s_{ij} = \underbrace{\sum_{i=1}^{k} \sum_{j=1}^{k} s_{ij}}_{= \frac{1}{2} k(k-1)} + \underbrace{\sum_{i=1}^{k} \sum_{j=k+1}^{n} s_{ij}}_{\leq k(n-k), \text{ as } s_{ij} \leq 1}$$

$$\leq \frac{1}{2} k(k-1) + k(n-k) = \sum_{i=1}^{k} (n-i).$$

Hence, $\mu_1 + \cdots + \mu_k \leq \lambda_{w(1)} + \cdots + \lambda_{w(k)} \leq \lambda_1 + \cdots + \lambda_k$, so $\mu \trianglelefteq \lambda$.

Thus, $R_\lambda(x;t)$ is a sum of elements of the form $(-t)^d \operatorname{sgn}(w)\, s_\mu$, $\mu \trianglelefteq \lambda$, $|\mu| = |\lambda|$.

We now compute the coefficient $u_{\lambda\lambda}(t)$. Indeed, we have $\lambda = \mu$ if l if $\lambda_{w(i)} = x_i$ and $s_{ij} = 1$ for all $i<j$. Hence,

$$u_{\lambda\lambda}(t) = \sum_{w \in S_n^\lambda} \operatorname{sgn}(w)\,(-t)^d,$$

with $$d = \sum_{i<j} r_{ji} = \sum_{i<j} s_{w^{-1}(j),w^{-1}(i)} = n(w^{-1}),$$

where $n(w^{-1}) = \#\{i<j : w^{-1}(j) < w^{-1}(i)\}$ is the number of inversions of w^{-1}.

Since $n(w) = n(w^{-1})$ and $\operatorname{sgn}(w) = (-1)^{n(w)}$, then

$$u_{\lambda\lambda}(t) = \sum_{w \in S_n^\lambda} t^{n(w)}.$$

<u>Lemma</u> $$\sum_{w \in S_m} t^{n(w)} = 1(1+t)\cdots(1+t+\cdots+t^{m-1}) = [m]_t!$$

Construct a bijection $I: \{0\}\times\{0,1\}\times\cdots\times\{0,\ldots,m-1\} \to S_m$ as follows. For $(s_1,\ldots,s_m) \in \{0\}\times\{0,1\}\times\cdots\times\{0,\ldots,m-1\}$, we set $I(s_1,\ldots s_m)$ to be the word in $1,\ldots m$ such that i is put in the exact position such that it precedes exactly s_i elements among $1,2,\ldots,i-1$. In particular, the placement of i creates s_i new inversions and we have $n(I(s_1,\ldots s_m)) = s_1+\cdots+s_m$,

Example: $m=5$.

$$I(0,0,2,1,2) = 31542$$

Thus, $$\sum_{w \in S_m} t^{n(w)} = \sum_{(S_{i_1} \to S_m)} t^{n(I(S_{i_1} \to S_m))} = \sum_{(S_{i_1} \to S_m)} t^{S_{i_1} + \cdots + S_m} = [m]_t!$$

Since $S_n^\lambda = \prod_{i \geq 0} S_{m_i}$, then

$$u_{\lambda\lambda}(t) = \sum_{w \in S_n^\lambda} t^{n(w)} = \sum_{(w_0, w_1, \ldots)} t^{n(w_0) + n(w_1) + \cdots}$$

$$= \prod_{i \geq 0} \sum_{w_i \in S_{m_i}} t^{n(w_i)} = \prod_{i \geq 0} [m_i]_t!$$

For convenience, set $v_\lambda(t) = \prod_{i \geq 0} [m_i]_t!$

Corollary: We have $\sum_{w \in S_n} w \cdot \left(\prod_{i<j} \frac{x_i - t x_j}{x_i - x_j} \right) = [n]_t$.

Proof. Take $\lambda = 0$ in the original def of R_λ. Then, the expansion of R_λ in terms of Schur functions has only one summand equal to $[n]_t$, as $\lambda = 0$ has n parts equal to zero.

Lemma: The polynomial $R_\lambda(x_1,\dots,x_n;t)$ is divisible by $v_\lambda(t)$ in $\mathbb{Z}[t]$.

Proof. Suppose that $\lambda_{i_1} = \cdots = \lambda_{i_m}$ and let $I = \{i_1,\dots,i_m\}$. Then, any $w \in S_I$ fixes $x_1^{\lambda_1}\cdots x_n^{\lambda_n}$, and by the corollary we get:

$$R_\lambda(x_1,\dots,x_n;t) = \sum_{w\in S_n/S_I}\sum_{w'\in S_I} ww'\cdot\left(x_1^{\lambda_1}\cdots x_n^{\lambda_n}\prod_{i<j}\frac{x_i-tx_j}{x_i-x_j}\right)$$

$$= [m]_t\cdot\sum_{w\in S_n/S_I} w\cdot\left(x_1^{\lambda_1}\cdots x_n^{\lambda_n}\prod_{\substack{i<j\\(i,j)\notin I\times I}}\frac{x_i-tx_j}{x_i-x_j}\right).$$

Proceeding in this way with the remaining parts of λ, we get:

$$R_\lambda(x_1,\dots,x_n;t) = v_\lambda(t)\sum_{w\in S_n/S_n^\lambda} w\cdot\left(x_1^{\lambda_1}\cdots x_n^{\lambda_n}\prod_{\lambda_i>\lambda_j}\frac{x_i-tx_j}{x_i-x_j}\right).$$

∎

Def We define the Hall-Littlewood polynomial $P_\lambda(x_1,\dots,x_n;t)$ as the quotient:

$$P_\lambda(x_1,\dots,x_n;t) = \frac{1}{v_\lambda(t)}R_\lambda(x_1,\dots,x_n;t)$$

$$= \frac{1}{v_\lambda(t)}\sum_{w\in S_n} w\cdot\left(x^\lambda\prod_{i<j}\frac{x_i-tx_j}{x_i-x_j}\right)$$

$$= \sum_{w\in S_n/S_n^\lambda} w\cdot\left(x^\lambda\prod_{\lambda_i>\lambda_j}\frac{x_i-tx_j}{x_i-x_j}\right).$$

Thus, $P_\lambda(x;t)$ is a symm. polynomial in $x_1,\dots,x_n$ with coefficients in $\mathbb{Z}[t]$.

Prop: Let $\ell(\lambda) \leq n$. Then, $P_\lambda(x_1, \dots x_n, 0, t) = P_\lambda(x_1, \dots, x_n, t)$.

Proof

We have $P_\lambda(x_1, \dots x_{n+1}, t) = \sum_{w \in S_{n+1}/S_{n+1}^\lambda} w \cdot \left(x_1^{\lambda_1} \cdots x_{n+1}^{\lambda_{n+1}} \prod_{i<j} \frac{x_i - t x_j}{x_i - x_j} \right)$.

Setting $x_{n+1} = 0$, the only summands for which the RHS are non-zero are those for which $w \in S_{n+1}$ permutes $n+1$ with r such that $\lambda_r = 0$, which is a subset of the permutations preserving the zero parts of λ. Taking w modulo S_{n+1}^λ, then w fixes $n+1$, ∎

We define the symmetric function $P_\lambda(x, t)$ in inf. many variables as the element of $\Lambda[t]$ whose image in $\Lambda_n[t]$ is $P_\lambda(x_1, \dots x_n, t)$ for every $n \geq \ell(\lambda)$.

The following properties of $P_\lambda(x, t)$ are immediate

- $P_\lambda(x_1, \dots, x_n, 0) = s_\lambda(x_1, \dots x_n)$
- $P_\lambda(x_1, \dots, x_n, 1) = m_\lambda(x_1, \dots, x_n)$
- $P_\lambda(x_1, \dots x_n, t) = \sum_{\mu \leq \lambda} w_{\lambda\mu}(t)\, s_\mu(x_1, \dots, x_n)$,

where $w_{\lambda\mu}(t) = 0$ unless $|\lambda| = |\mu|$ and $\mu \leq \lambda$, with $w_{1\lambda}(t) = 1$.

- $P_{(1^r)}(x_1, \dots, x_n, t) = e_r(x_1, \dots x_n)$

In particular, the transition matrix which expresses P_λ in terms of the s_μ is strictly upper triangular. Since the s_μ form a $\mathbb{Z}[t]$-basis of $\Lambda[t]$, we have:

- The symm. functions $P_\lambda(x;t)$ form a $\mathbb{Z}[t]$-basis of $\Lambda[t]$.

Hence one may write the Schur function s_λ in terms of the $P_\mu(x,t)$ as follows:

$$s_\lambda(x_1,\dots,x_n) = \sum_{\mu \trianglelefteq \lambda} K_{\lambda\mu}(t)\, P_\mu(x_1,\dots,x_n,t),$$

where $K_{\lambda\mu}(t)$ are t-analogues of the Kostka numbers and $K_{\lambda\mu}(t)$ are known as the Kostka-Foulkes polynomials.

2. Hall algebras

Let G be an abelian p-group. Then,

$$G \simeq \mathbb{Z}/p^{\lambda_1} \oplus \cdots \oplus \mathbb{Z}/p^{\lambda_n},$$

where $\lambda = (\lambda_1, \ldots, \lambda_n)$ is some partition. We say G is of <u>type λ</u>.

We may view G as a $\mathbb{Z}$-module. In particular, $n \cdot G$ means the submodule generated by all elements of the form $n \cdot x$.

<u>Lemma</u> Let G be of type λ. Then,

$$G \supseteq pG \supseteq p^2 G \supseteq \cdots \supseteq p^{|\lambda|} G = \{0\}$$

with $p^{i-1}G/p^iG$ being of type $(1^{\lambda_i'})$, where $\lambda' = (\lambda_1', \ldots \lambda_s')$ is the transpose.

<u>Proof</u> Let $\langle x_j \rangle = \mathbb{Z}/p^{\lambda_j}\mathbb{Z}$. Then, $\lambda_j \geq i$ if/if $p^{i-1}\langle x_j \rangle$ is nontrivial if/if $p^i \langle x_j \rangle \leq p^{i-1}\langle x_j \rangle$ is proper if/if $p^{i-1}\langle x_j \rangle / p^i \langle x_j \rangle \simeq \mathbb{Z}/p\mathbb{Z}$.

Thus, $$p^{i-1}G/p^iG \simeq \bigoplus_{\lambda_j \geq i} \mathbb{Z}/p\mathbb{Z}$$

Since $\lambda_i' = \{j : \lambda_j \geq i\}$, the result follows.

∎

For partitions $\mu^{(1)}, \cdots, \mu^{(r)}, \lambda$, let

$$G^{\lambda}_{\mu^{(1)}, \cdots, \mu^{(r)}}$$

be the number of chain of subgroups of a type λ group G:

$$G = G_0 \underbrace{\supseteq G_1 \supseteq \cdots}_{\mu^{(1)}} \underbrace{\supseteq G_r}_{\mu^{(r)}} = \{0\}$$

such that G_{i-1}/G_i is of type $\mu^{(i)}$.

In particular, $G^{\lambda}_{\mu\nu}$ is the number of subgroups N of type ν with G/N of type μ (cotype):

$$G \underbrace{\supseteq N}_{\mu} \overbrace{\supseteq \{0\}}^{\nu}$$

Let H be a free $\mathbb{Z}$-module on a basis (u_λ) indexed by all partitions λ. Define a product on H as follows:

$$u_\mu \cdot u_\nu = \sum_\lambda G^{\lambda}_{\mu\nu} u_\lambda$$

Since $G^{\lambda}_{\mu\nu} = 0$ unless $|\lambda| = |\mu| + |\nu|$, the above sum is finite.

Prop H is a commutative and associative ring with identity, where the identity is u_0 (zero partition).

The ring H is called the (classical) Hall algebra.

Prop The Hall algebra H is generated as a $\mathbb{Z}$-algebra by the elements $u_{(1^r)}$ and they are algebraically independent over $\mathbb{Z}$.

Proof. Let λ be a partition with conjugate $\lambda' = (\lambda_1', \dots, \lambda_s')$. Consider the product over the columns of λ:

$$\pi_\lambda = u_{(1^{\lambda_1'})} \cdot u_{(1^{\lambda_2'})} \cdots u_{(1^{\lambda_s'})}.$$

Then, π_λ can be written as a linear comb. of the u_μ:

$$\pi_\lambda = \sum_\mu a_{\lambda\mu} u_\mu,$$

where $a_{\lambda\mu}$ is the number of filtrations of a type μ group

$$G = G_0 \supseteq G_1 \supseteq \cdots \supseteq G_s = \{0\}$$

such that $G_{i-1}/G_i \cong \mathbb{Z}/p\mathbb{Z}^{\oplus \lambda_i'}$ is of type $(1^{\lambda_i'})$.

If $a_{\lambda\mu} \neq 0$, then $p \cdot G_{i-1} \subseteq G_i \ \forall i$, whence $p^i G \subseteq G_i$. Thus,

$$|G/G_i| \leq |G/p^i G|.$$

Since $G/G_1 \simeq G/G_2 \Big/ G_1/G_2$, then

$$|G/G_2| = |G/G_1| \cdot |G_1/G_2| = p^{\lambda_1' + \lambda_2'}.$$

By induction, we find that $|G/G_i| = p^{\lambda_1' + \cdots + \lambda_i'}$. Similarly, one finds that $|G/p^iG| = p^{\mu_1' + \cdots + \mu_i'}$ from the lemma. Hence,

$$|G/G_i| \leq |G/p^iG| \Rightarrow \lambda_1' + \cdots + \lambda_i' \leq \mu_1' + \cdots + \mu_i'$$

Thus, $\lambda' \leq \mu'$ implies $\mu \leq \lambda$.

Now, if $\mu = \lambda$, then the only possible chain is

$$G \supseteq pG \supseteq p^2G \supseteq \cdots \supseteq \{0\},$$

because of the condition $p^iG \subseteq G_i$. Thus, we find that $a_{\lambda\mu} = 0$, unless $\mu \leq \lambda$ and $a_{\lambda\lambda} = 1$.

Therefore, the matrix $(a_{\lambda\mu})$ is strictly upper unitriangular, whence we can solve for the u_μ as integral linear combinations of the u_{π_λ}'s.

Moreover, the u_{π_λ}'s are $\mathbb{Z}$-linear independent, whence the elements $u_{(1^r)}$ are algebraically independent over $\mathbb{Z}$.

$\square$

As a result, the Hall algebra H and the ring Λ of symmetric functions are isomorphic.

By viewing both H and Λ as $\mathbb{Q}$-algebras, an interesting choice of isomorphism is the following $\mathbb{Q}$-linear map:

$$\begin{aligned} \psi : H &\longrightarrow \Lambda \\ u_{(1^r)} &\mapsto p^{-\frac{r(r-1)}{2}} \cdot e_r(x) . \end{aligned}$$

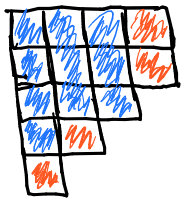
We shall see that the Hall-Littlewood polynomial $P_\lambda(x, p^{-1})$ appears in the image of u_λ through ψ.

3. The isomorphism $\psi: H \to \Lambda$.

We develop some language that will be useful.

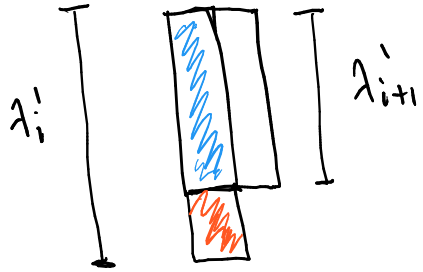
Def Let $\lambda \supseteq \mu$ be partitions. We say that $\lambda - \mu$ is a vertical m-strip if $\lambda_i - \mu_i \leq 1$ and $\sum \lambda_i - \mu_i = m$.

Example:



Lemma: $\lambda - \mu$ is a vertical m-strip if f if $\mu_i' \geq \lambda_{i+1}'$ iff if $\lambda_i' - \mu_i' \leq m_{i-1}^{\mu}$, where $m_{i-1}^{\mu} = \#$ parts of μ of size $i-1$.

Proof: The first equivalence is visually clear:



Now, notice that $m_{i-1}^{\mu} = \#\{j : \mu_j \geq i-1\} - \#\{j : \mu_j \geq i\}$

$= \mu_{i-1}' - \mu_i'$. Hence,

$$\mu_{i-1}' \geq \lambda_i' \text{ if f if } m_{i-1}^{\mu} = \mu_{i-1}' - \mu_i' \geq \lambda_i' - \mu_i'.$$

∎

Prop: Let μ be a partition, $k = \mu_1$, $n \geq \ell(\mu) + m$. Then, there is a bijection:

$$\left\{ r = (r_0, \dots, r_k) \in \mathbb{N}^{k+1} : r_i \leq m_i^\mu, \sum r_i = m \right\} \longleftrightarrow \left\{ \lambda : \lambda \supseteq \mu, \lambda - \mu \text{ is a vertical } m\text{-strip} \right\}$$

Proof: For each $r = (r_0, \dots, r_k)$, we construct λ by adding r_{i-1} boxes to the i-th column of μ. The bijectivity follows from the previous lemma. ∎

Rmk: We think of the set $\{ r = (r_0, \dots, r_k) : r_i \leq m_i^\mu, \sum r_i = m \}$ as all the possible ways to fill with boxes the subsequent gaps $m_i^\mu = \mu_i' - \mu_{i+1}'$.

We now turn to the Hall-Littlewood polynomials again.

The product of Hall-Littlewood polynomials is a $\mathbb{Z}[t]$-linear combination of the $P_\lambda(x;t)$:

$$P_\mu(x;t) P_\nu(x;t) = \sum_\lambda f_{\mu\nu}^\lambda(t) \, P_\lambda(x;t),$$

where the sum is over λ with $|\lambda| = |\mu| + |\nu|$.

In particular, we have that $f_{\mu\nu}^\lambda(0) = c_{\mu\nu}^\lambda$ is the Littlewood-Richardson coefficients of the expansion of $s_\mu(x) s_\nu(x)$.

Define the t-analogue of the binomial coefficient:

$$\begin{bmatrix} n \\ r \end{bmatrix}_t = \begin{cases} \dfrac{[n]_t!}{[n-r]_t! \, [r]_t!}, & 0 \leq r \leq n \\ 0, & \text{else.} \end{cases}$$

Prop: We have

$$f^{\lambda}_{\mu(1^m)}(t) = \prod_{i\geq 1} \begin{bmatrix} \lambda_i' - \lambda_{i+1}' \\ \lambda_i' - \mu_i' \end{bmatrix}_t$$

so that $f^{\lambda}_{\mu(1^m)}(t) = 0$, unless $\lambda - \mu$ is a vertical m-strip.

Proof. Set $n \geq \ell(\mu) + m$ (this condition is compatible with the Littlewood-Richardson rule: $\lambda - \mu$ has weight 1^m).

Let $K = \mu_1$. We split $\{x_1, \dots, x_n\}$ into disjoint subsets $X_0, \dots, X_K$, where in X_i we collect all the variables x_j such that $\mu_j = i$, so that $|X_i| = m_i^{\mu}$. Write $X_i = \{x_1^{(i)}, \dots, x_{m_i^{\mu}}^{(i)}\}$. Then,

$$\prod_{i=1}^{n} (1 + x_i t) = \prod_{i=0}^{K} \prod_{\ell=1}^{m_i^{\mu}} (1 + x_\ell^{(i)} t)$$

implies:

$$e_m(x_1, \dots, x_n) = \sum_{\substack{r=(r_0,\dots,r_K) \\ r_i \leq m_i^{\mu}}} e_{r_0}(X_0)\, e_{r_1}(X_1) \cdots e_{r_K}(X_K)$$

Pick a summand above indexed by some $r = (r_0, \dots, r_K)$, $r_i \leq m_i^{\mu}$, and let $\lambda \supseteq \mu$ be the corresponding partition with $\lambda - \mu$ being a m-strip under the bijection we saw previously: $\lambda_i' - \mu_i' = r_{i-1}$.

Since $e_{r_i}(X_i) = P_{(1^{r_i})}(X_i, t) = \frac{1}{v_{(1^{r_i})}} \sum_{w_i \in S_{m_i}} w_i \cdot \left(x_1^{(i)} \cdots x_{r_i}^{(i)} \prod_{a<b} \frac{x_a^{(i)} - t x_b^{(i)}}{x_a^{(i)} - x_b^{(i)}} \right)$,

then we have:

$$e_{r_0}(X_0) \cdots e_{r_k}(X_k) = \frac{1}{c_{\lambda,\mu}(t)} \Phi_{\lambda,\mu}(x_1, \ldots, x_n, t),$$

where $c_{\lambda,\mu}(t) = v_{(1^{r_0})}(t) \cdots v_{(1^{r_k})}(t) = \prod_{i=0}^{k} [r_i]_t! \, [m_i^{\mu} - r_i]_t!$ and

$$\Phi_{\lambda,\mu}(x,t) = \sum_{(w_0, w_1, \ldots, w_k) \in S_n^{\mu}} w_0 \cdot \left(x_1^{(0)} \cdots x_{r_0}^{(0)} \prod_{i<j} \frac{x_i^{(0)} - t x_j^{(0)}}{x_i^{(0)} - x_j^{(0)}} \right) \cdots w_k \cdot \left(x_1^{(k)} \cdots x_{r_k}^{(k)} \prod_{i<j} \frac{x_i^{(k)} - t x_j^{(k)}}{x_i^{(k)} - x_j^{(k)}} \right).$$

Note: $x_1^{(0)} \cdots x_{r_0}^{(0)} \, x_1^{(1)} \cdots x_{r_1}^{(1)} \cdots x_1^{(k)} \cdots x_{r_k}^{(k)} = x_1^{\lambda_1 - \mu_1} \cdots x_n^{\lambda_n - \mu_n}$,

(because $r_i = 0$ implies $\lambda_j - \mu_j = 0$ for every j such that $\lambda_j = i$)

whence $\Phi_{\lambda,\mu}(x,t) = \sum_{w \in S_n^{\mu}} w \cdot \left(x_1^{\lambda_1 - \mu_1} \cdots x_n^{\lambda_n - \mu_n} \prod_{\substack{i<j \\ \mu_i = \mu_j}} \frac{x_i - t x_j}{x_i - x_j} \right)$.

By summing over all such λ such that $\lambda - \mu$ is a vertical n-strip, we can write

$$e_m(x_1, \ldots, x_n) = \sum_{\lambda} \frac{1}{c_{\lambda,\mu}(t)} \Phi_{\lambda,\mu}(x_1, \ldots, x_n, t).$$

Since $P_\mu(x;t) = \sum_{w \in S_n/S_n^\mu} w \cdot \left(x_1^{\mu_1} \cdots x_n^{\mu_n} \prod_{\lambda_i > \lambda_j} \frac{x_i - t x_j}{x_i - x_j} \right)$ and $e_m(x)$ is symmetric, then:

$$P_\mu(x;t)\, e_m(x) = \sum_{w \in S_n/S_n^\mu} w \cdot \left(x_1^{\mu_1} \cdots x_n^{\mu_n} \prod_{\lambda_i > \lambda_j} \frac{x_i - t x_j}{x_i - x_j}\, e_m(x) \right)$$

$$= \sum_{w \in S_n/S_n^\mu} w \cdot \left(\sum_{\lambda} \frac{1}{c_{\lambda,\mu}(t)}\, x_1^{\mu_1} \cdots x_n^{\mu_n} \prod_{\lambda_i > \lambda_j} \frac{x_i - t x_j}{x_i - x_j}\, \Phi_{\lambda,\mu}(x;t) \right).$$

Since the term $x_1^{\mu_1} \cdots x_n^{\mu_n} \prod_{\lambda_i > \lambda_j} \frac{x_i - t x_j}{x_i - x_j}$ is symmetric with respect to S_n^μ, then from the definition of $\Phi_{\lambda,\mu}(x;t)$ we get:

$$x_1^{\mu} \cdots x_n^{\mu_n} \prod_{\lambda_i > \lambda_j} \frac{x_i - t x_j}{x_i - x_j}\, \Phi_{\lambda,\mu}(x;t) = \sum_{w' \in S_n^\mu} w' \cdot \left(x_1^{\lambda_1} \cdots x_n^{\lambda_n} \prod_{i<j} \frac{x_i - t x_j}{x_i - x_j} \right).$$

Therefore, we see that:

$$P_\mu(x;t)\, e_m(x) = \sum_{\lambda} \frac{1}{c_{\lambda,\mu}(t)} \sum_{w \in S_n/S_n^\mu} w \cdot \sum_{w' \in S_n^\mu} w' \cdot \left(x_1^{\lambda_1} \cdots x_n^{\lambda_n} \prod_{i<j} \frac{x_i - t x_j}{x_i - x_j} \right)$$

$$= \sum_{\lambda} \frac{1}{c_{\lambda,\mu}(t)}\, R_\lambda(x;t),$$

the sum being over λ such that $\lambda - \mu$ is a vertical m-strip.

Since $P_{(1^m)}(x,t) = e_m(x)$ and $R_\lambda(x,t) = \sigma_\lambda(t)\, P_\lambda(x,t)$, then

$$f^{\lambda}_{\mu(1^m)}(t) = \frac{\sigma_\lambda(t)}{c_{\lambda,\mu}(t)} = \frac{\prod_{i=0}^{\lambda_1} [m_i^\lambda]_t!}{\prod_{i=0}^{\mu_1} [r_i]_t!\,[m_i^\mu - r_i]_t!}$$

Since $m_i^\lambda = \lambda_i' - \lambda_{i+1}'$, $m_i^\mu = \mu_i' - \mu_{i+1}'$ and $r_{i-1} = \lambda_i' - \mu_i'$, then

$$\frac{[m_i^\lambda]_t!}{[m_i^\mu - r_i]_t!\,[r_i]_t!} = \frac{[\lambda_i' - \lambda_{i+1}']_t!}{[\mu_i' - \lambda_{i+1}']_t!\,[\lambda_{i+1}' - \mu_{i+1}']_t!}.$$

Taking the product as above and cancelling a few terms, we get:

$$f_{\mu(1^m)}(t) = \prod_{i\geq 1} \frac{[\lambda_i' - \lambda_{i+1}']_t!}{[\mu_i' - \lambda_{i+1}']_t!\,[\lambda_i' - \mu_i']_t!} = \prod_{i\geq 1} \begin{bmatrix} \lambda_i' - \lambda_{i+1}' \\ \lambda_i' - \mu_i' \end{bmatrix}$$

□

The following result on the structural constants of the Hall algebra will be assumed without proof.

For λ partition, let $n(\lambda) = \sum_{i \geq 1} (i-1)\lambda_i$. In particular, $n(1^r) = \frac{r(r-1)}{2}$.

Thm: We have: $$G^{\lambda}_{\mu\,(1^m)} = p^{n(\lambda) - n(\mu) - n(1^m)} \prod_{i \geq 1} \begin{bmatrix} \lambda_i' - \lambda_{i+1}' \\ \lambda_i' - \mu_i' \end{bmatrix}_{p^{-1}}$$ ☒

Thus, it is immediate that:

$$G^{\lambda}_{\mu\,(1^m)} = p^{n(\lambda) - n(\mu) - n(1^m)} f_{\mu\,(1^m)}(p^{-1}).$$

Thm Let $\psi : H \to \Lambda$ be the isomorphism of $\mathbb{Q}$-algebras:

$$u_{(1^r)} \mapsto p^{-n(1^r)} e_r(x).$$

Then, $\psi(u_\lambda) = p^{-n(\lambda)} P_\lambda(x, p^{-1})$.

Proof. Since the $P_\lambda(x, p^{-1})$ form a $\mathbb{Q}$-basis of Λ, we may define another isomorphism of $\mathbb{Q}$-algebras by

$$\psi' : H \to \Lambda$$
$$u_\lambda \mapsto p^{-n(\lambda)} P_\lambda(x, p^{-1})$$

and prove that $\psi = \psi'$.

We show that $\psi(u_\lambda) = \psi'(u_\lambda)$ by induction on λ along the partial order $\trianglelefteq$.

For $\lambda = 0$ the zero-partition, we clearly have $\psi(u_0) = \psi'(u_0) = 1$. Now, we assume $\lambda \neq 0$ and let μ be the partition obtained from λ by deleting the last column with $\lambda'_\kappa = m$, $\kappa = \lambda_1$. In particular, we have $\mu < \lambda$ and

$$n(\lambda) = \sum_{i=1}^{m} (i-1)\underbrace{\lambda_i}_{\mu_i+1} + \sum_{i=m+1}^{n} (i-1)\underbrace{\lambda_i}_{\mu_i} = \sum_{i=1}^{n} (i-1)\mu_i + \sum_{i=1}^{m} (i-1) = n(\mu) + n(1^m).$$

Moreover, since $\lambda'_i = \mu'_i$ for $1 \leq i < 1$, then

$$f^{\lambda}_{\mu(1^m)}(t) = \prod_{i \geq 1} \begin{bmatrix} \lambda'_i - \lambda'_{i+1} \\ \lambda'_i - \mu'_i \end{bmatrix} = 1.$$

Thus, $G^{\lambda}_{\mu(1^m)} = 1$. Since $f^{\nu}_{\mu(1^m)}(t) = 0$ unless $\nu - \mu$ is a vertical m-strip, the same is true for $G^{\nu}_{\mu(1^m)}$. In such cases we also have $\nu < \lambda$. Hence,

$$u_\mu u_{(1^m)} = u_\lambda + \sum_{\nu < \lambda} G^{\nu}_{\mu(1^m)} u_\nu. \qquad (*)$$

By the same reasoning, we can write:

$$P_\mu(x, p^{-1}) P_{(1^m)}(x, p^{-1}) = P_\lambda(x, p^{-1}) + \sum_{\nu < \lambda} f^{\nu}_{\mu(1^m)}(p^{-1}) P_\nu(x, p^{-1}). \qquad (**)$$

We apply ψ on $(*)$ along with the induction hyphotesis to get:

$$P_\mu(x, p^{-1})\, e_m(x)\, p^{-n(\mu)-n(1^m)} = \psi'(u_\lambda) + \sum_{\nu<\lambda} G^{\nu}_{\mu(1^m)}\, p^{-n(\nu)}\, P_\nu(x, p^{-1}).$$

Since $G^{\nu}_{\mu(1^m)}\, p^{-n(\nu)} = p^{-n(\mu)-n(1^m)}\, f^{\nu}_{\mu(1^m)}(p^{-1})$, the above reduces to:

$$P_\mu(x, p^{-1})\, P_{(1^m)}(x, p^{-1}) = p^{n(\mu)+n(1^m)}\, \psi(u_\lambda) + \sum_{\nu<\lambda} f^{\nu}_{\mu(1^m)}(p^{-1})\, P_\nu(x, p^{-1}).$$

Since $n(\lambda) = n(\mu) + n(1^m)$, by comparing with $(**)$ we get

$$p^{n(\lambda)}\, \psi'(u_\lambda) = P_\lambda(x, p^{-1}) \Rightarrow \psi'(u_\lambda) = p^{-n(\lambda)}\, P_\lambda(x, p^{-1}).$$

$\square$